

MEMBER 4: FRONTEND DEVELOPER - VIBE CODING ROADMAP

Smart India Hackathon 2026 | Team ILLUMINATI (ID: 27113) | PS: SIH1642

Primary Folder: frontend/ | Dedicated Branch: feature/react-dashboard | Deliverables: 8 Components

PHASE 1: SETUP & COMMAND-CENTER DESIGN SYSTEM

Prompt 4.1

Initialize Vite + React in frontend/. Install Tailwind CSS, lucide-react, recharts, leaflet, react-leaflet, and axios. Configure tailwind.config.js with dark slate theme and severity colors: Normal (#10b981), Watch (#f59e0b), High (#f97316), Critical (#ef4444).

Prompt 4.2

Create frontend/src/services/api.js with Axios pointing to http://localhost:8000/api/v1. Include local mock fallback responses for every endpoint so frontend can be developed completely without running backend.

Prompt 4.3

Create frontend/src/components/Navbar.jsx with AeroSense-AI brand logo, Team ILLUMINATI badge, live ticking clock, API status pill, and navigation tabs (National Map, Station Deep-Dive, What-If Studio, Historical Trends).

Prompt 4.4

Create frontend/src/components/ExecutiveSummaryCards.jsx displaying 4 metric counters: Total Stations Monitored, Critical Alerts, High Alerts, and Stations Normal with glowing status indicators.

PHASE 2: INTERACTIVE INDIA LEAFLET MAP

Prompt 4.5

Create frontend/src/components/IndiaAnomalyMap.jsx using react-leaflet centered on India (lat: 20.59, lon: 78.96, zoom: 5) with dark CartoDB tiles.

Prompt 4.6

In IndiaAnomalyMap.jsx, create custom pulsating SVG marker icons matching station severity: Green (Normal), Yellow (Watch), Orange (High), Red (Critical).

Prompt 4.7

In IndiaAnomalyMap.jsx, bind Leaflet Popups to pins showing city name, current temperature, rainfall, anomaly score %, and severity badge.

Prompt 4.8

In IndiaAnomalyMap.jsx, add onSelectLocation(cityName) callback when a user clicks a pin, switching the active deep-dive panel to that city.

PHASE 3: STATION DEEP-DIVE & BASELINE GAUGES

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Prompt 4.9

Create frontend/src/components/StationDeepDive.jsx showing selected city name, coordinates, elevation, and overall anomaly status banner.

Prompt 4.10

In StationDeepDive.jsx, create 5 metric gauge cards: Temperature, Rainfall, Humidity, Pressure, Wind Speed. Show Observed Value, Expected Seasonal Normal, and Deviation Delta (color-coded red for +9.9 deg C).

Prompt 4.11

In StationDeepDive.jsx, add a visual progress bar or mini-corridor for each metric showing where observed reading sits relative to normal $\mu \pm 2\sigma$ corridor.

PHASE 4: WHAT-IF PREDICTION STUDIO (SIH DEMO FEATURE)

Prompt 4.12

Create frontend/src/components/PredictionStudio.jsx with city selector (default: Bengaluru) and month selector.

Prompt 4.13

In PredictionStudio.jsx, create 5 interactive sliders with numeric inputs: Temperature (10-50 deg C), Rainfall (0-200 mm), Humidity (10-100%), Pressure (970-1040 hPa), Wind Speed (0-40 m/s).

Prompt 4.14

In PredictionStudio.jsx, add quick-preset buttons for SIH demo: 'Normal September Day', 'Extreme Heatwave', 'Bengaluru Cloudburst (145mm Rain + 994 hPa)', 'Cyclone Depression'. Clicking populates sliders instantly.

Prompt 4.15

In PredictionStudio.jsx, trigger debounced POST /predict on slider changes. Show a large gauge with Anomaly Score (0-100%), animated severity badge (CRITICAL/HIGH/WATCH/NORMAL), and anomaly type tag.

PHASE 5: EXPLAINABILITY & HISTORICAL TRENDS

Prompt 4.16

Create frontend/src/components/ExplainabilityCard.jsx displaying natural-language diagnostic alert string in a high-contrast banner.

Prompt 4.17

In ExplainabilityCard.jsx, use Recharts BarChart to render a horizontal bar chart ranking contributor variables by contribution percentage (e.g. Rainfall 52.4%, Temp 28.1%, Pressure 19.5%).

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Prompt 4.18

Create `frontend/src/components/HistoricalTrends.jsx`. Use Recharts AreaChart & LineChart to plot 30-day time-series: shaded green area for Normal Corridor ($\mu \pm 2\sigma$), blue line for observed readings, red dots for breaches.

Prompt 4.19

In `HistoricalTrends.jsx`, add seasonal anomaly frequency bar chart showing occurrences in Winter, Summer, Monsoon, and Post-Monsoon.

Prompt 4.20

In `frontend/src/App.jsx`, wire all components into a sleek, responsive dark layout. Add toast notification when anomaly escalates to Critical. Ensure layout works seamlessly on laptops and projectors.